Computer Networks

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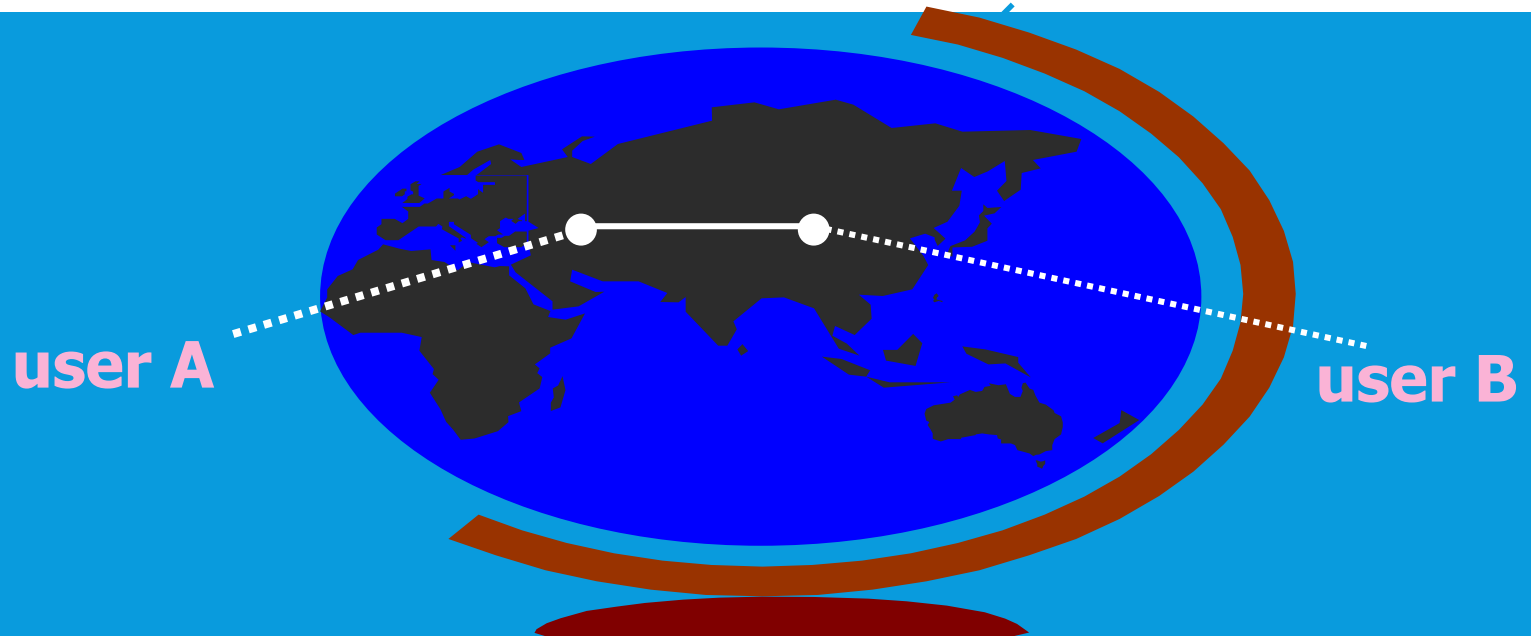
Lecture # 24

Link Layer

Reliable Transmission Over Unreliable Physical Link



Functions of the Link Layer

- Provides service interface to the network layer
- Dealing with transmission errors
- Regulating data flow: Slow receivers not swamped by fast senders

- To accomplish these goals, the link layer encapsulates the packets into **frames**.

FRAMING

- Flag bytes with byte/character stuffing
  - Each frame starts and ends with special bytes – flag byte
  - If the flag byte occurs in the frame, stuff an extra escape byte (ESC).
  - Used in PPP (Point to Point Protocol)
- Starting and ending flags, with bit stuffing
  - Each frame begins and ends with a special bit pattern 01111110 whenever the sender sees 5 consecutive 1s, it stuffs a 0.
- Physical layer coding violations
  - Only applicable to networks in which the encoding on the physical medium contains some redundancy.
  - For example, a 1 bit is a high-low pair and a 0 bit is a low-high pair. High-high and low-low are used for delimiter.

- Error Control
  - Detect and/or correct errors
  - Ensure that all frame are eventually delivered in order → use acknowledgement, timer, and sequence number
- Flow Control
  - Slow down the sender when the data is coming too fast for the receiver
  - Two approaches:
    - Feedback-based flow control – the receiver sends back permission to the sender.
    - Rate-based flow control – built-in mechanism that limits senders' rate. Rarely used in the data link layer.

- Error-correcting codes/forward error correction
  - include enough redundant information to enable the receiver to deduce the correct transmitted data.
  - Used in unreliable channel such as wireless links
- Error-detecting codes
  - include only enough redundancy to allow the receiver to request a retransmission.
  - Used in reliable channel such as fiber

Note

Data can be corrupted during transmission.

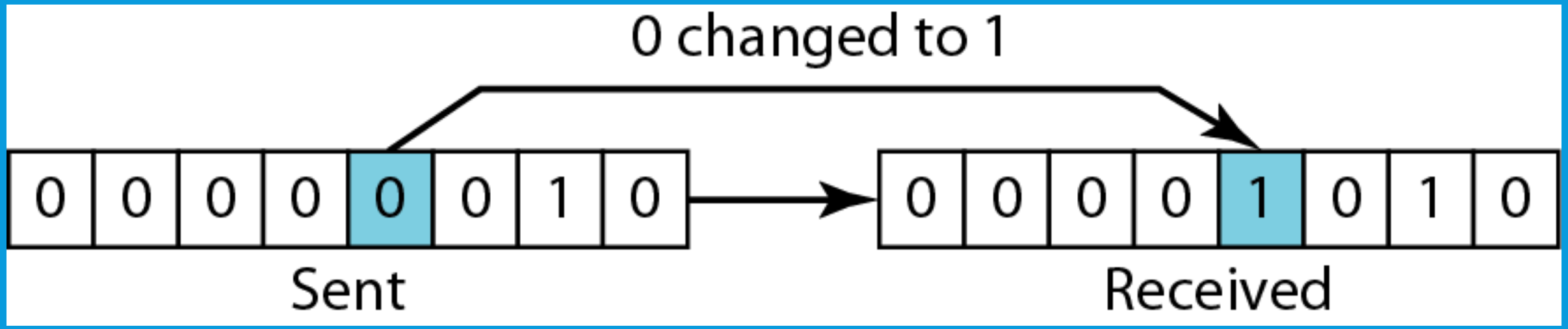
Some applications require that errors be detected and corrected.

Let us first discuss some issues related, directly or indirectly, to error detection and correction.

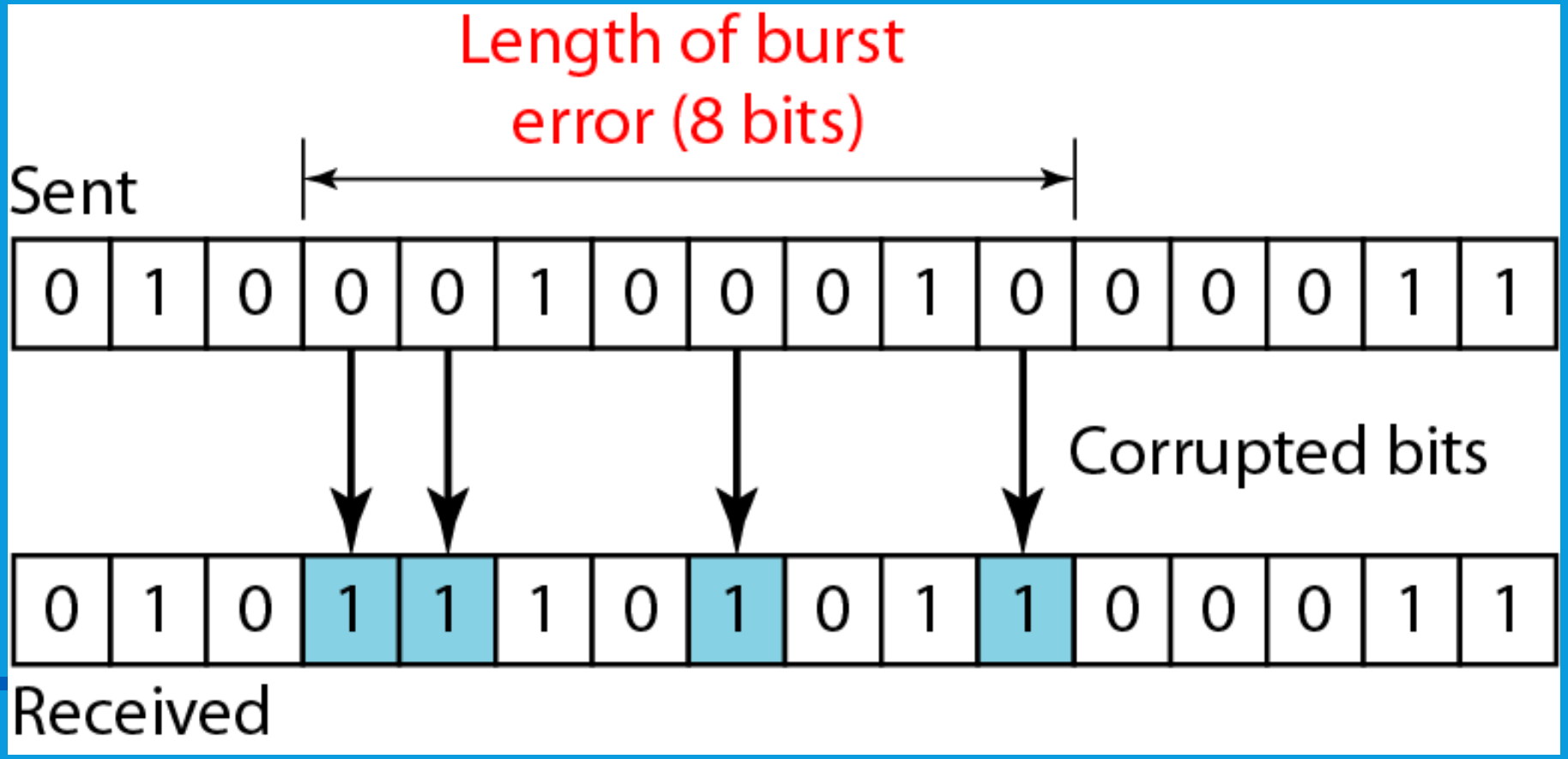
Topics to be discussed

- Types of Errors
- Redundancy
- Detection Versus Correction
- Forward Error Correction Versus Retransmission
- Coding
- Modular Arithmetic

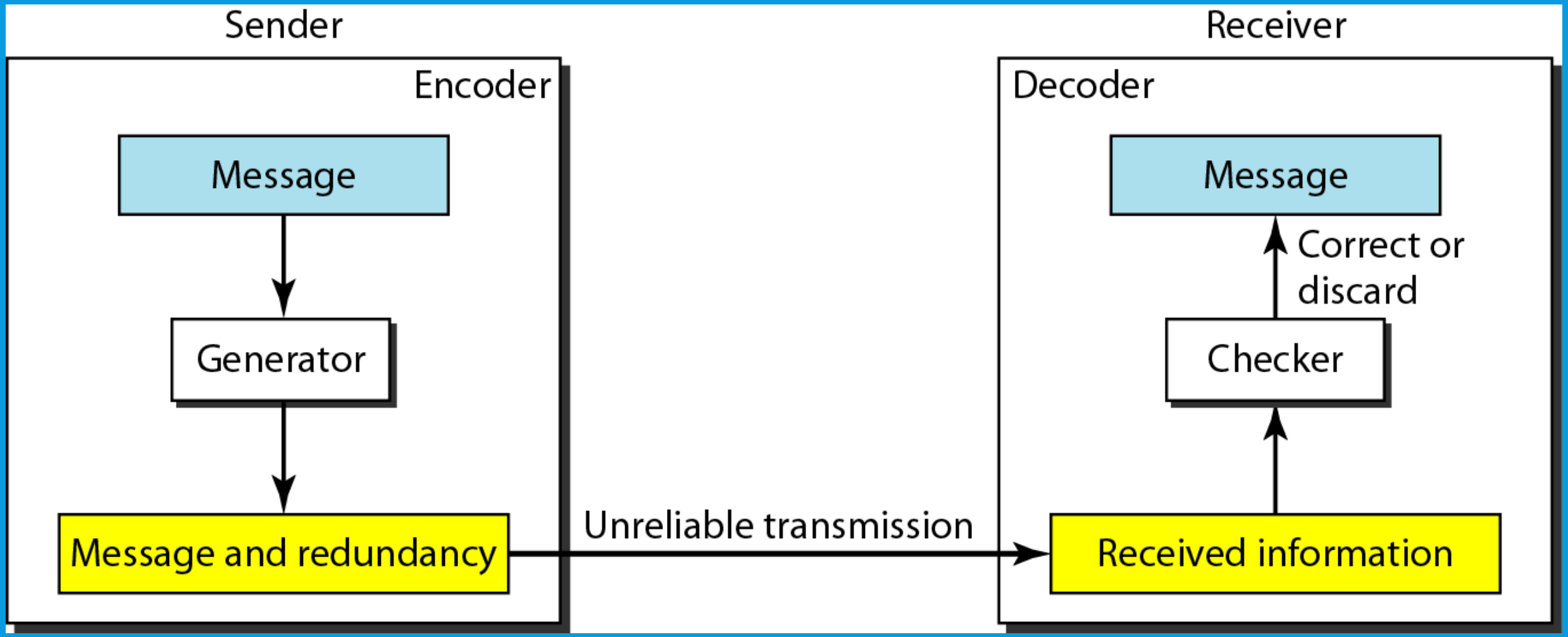
In a single-bit error, only 1 bit in the data unit has changed.



A burst error means that 2 or more bits in the data unit have changed.



To detect or correct errors, we need to send extra (redundant) bits with data.



In modulo-N arithmetic, we use only the integers in the range 0 to N –1, inclusive.

$0 \oplus 0 = 0$ $\quad$ $1 \oplus 1 = 0$

a. Two bits are the same, the result is 0.

$0 \oplus 1 = 1$ $\quad$ $1 \oplus 0 = 1$

b. Two bits are different, the result is 1.

$$
\begin{array}{cccccc}
 & 1 & 0 & 1 & 1 & 0 \\
\oplus & 1 & 1 & 1 & 0 & 0 \\
\hline
 & 0 & 1 & 0 & 1 & 0
\end{array}
$$

c. Result of XORing two patterns

In block coding, we divide our message into blocks, each of k bits, called ***datawords****. We add r redundant bits to each block to make the length n = k + r. The resulting n-bit blocks are called* ***codewords****.*

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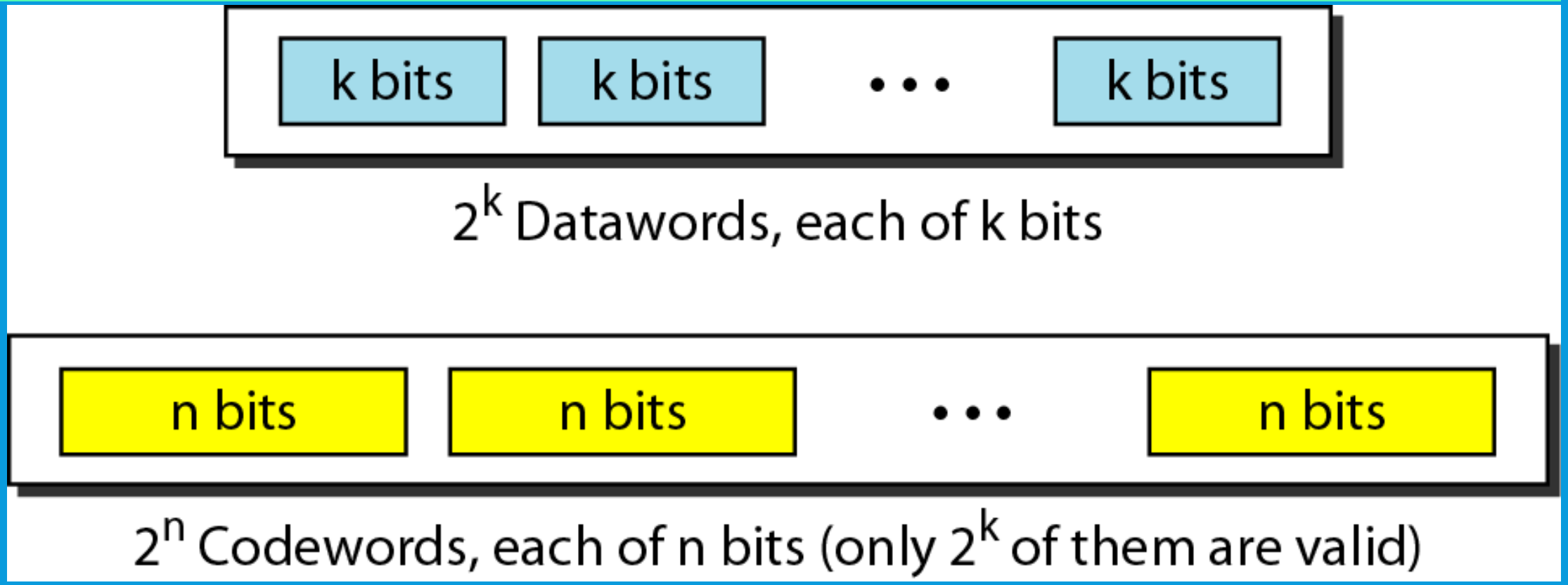
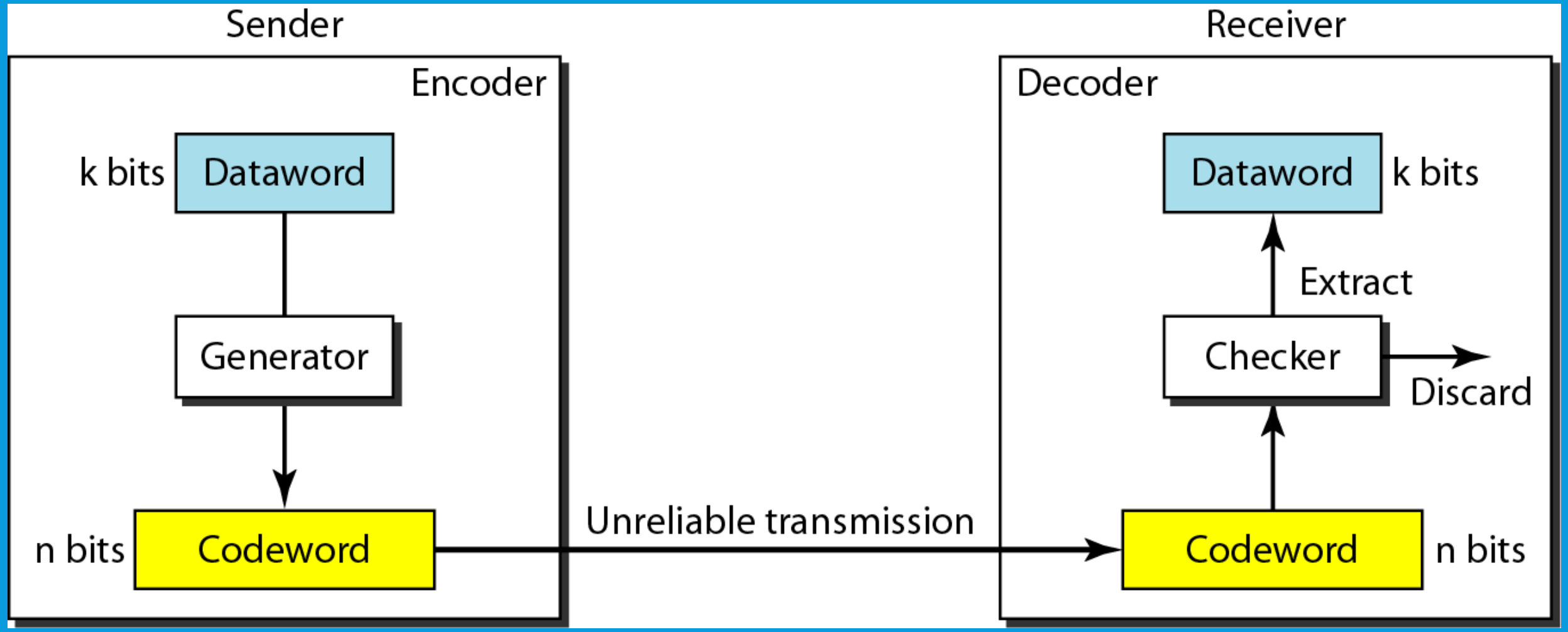


Datawords and codewords in block coding

The 4B/5B block coding is a good example of this type of coding. In this coding scheme, $k = 4$ and $n = 5$. As we saw, we have $2^k = 16$ datawords and $2^n = 32$ codewords. We saw that 16 out of 32 codewords are used for message transfer and the rest are either used for other purposes or unused.

ERROR DETECTION

- Enough redundancy is added to detect an error.
- The receiver knows an error occurred but does not know which bit(s) is(are) in error.
- Has less overhead than error correction.



Process of error detection in block coding

Let us assume that k = 2 and n = 3. Table shows the list of datawords and codewords. Later, we will see how to derive a codeword from a dataword.

Assume the sender encodes the dataword 01 as 011 and sends it to the receiver. Consider the following cases:

1. *The receiver receives 011. It is a valid codeword. The receiver extracts the dataword 01 from it.*

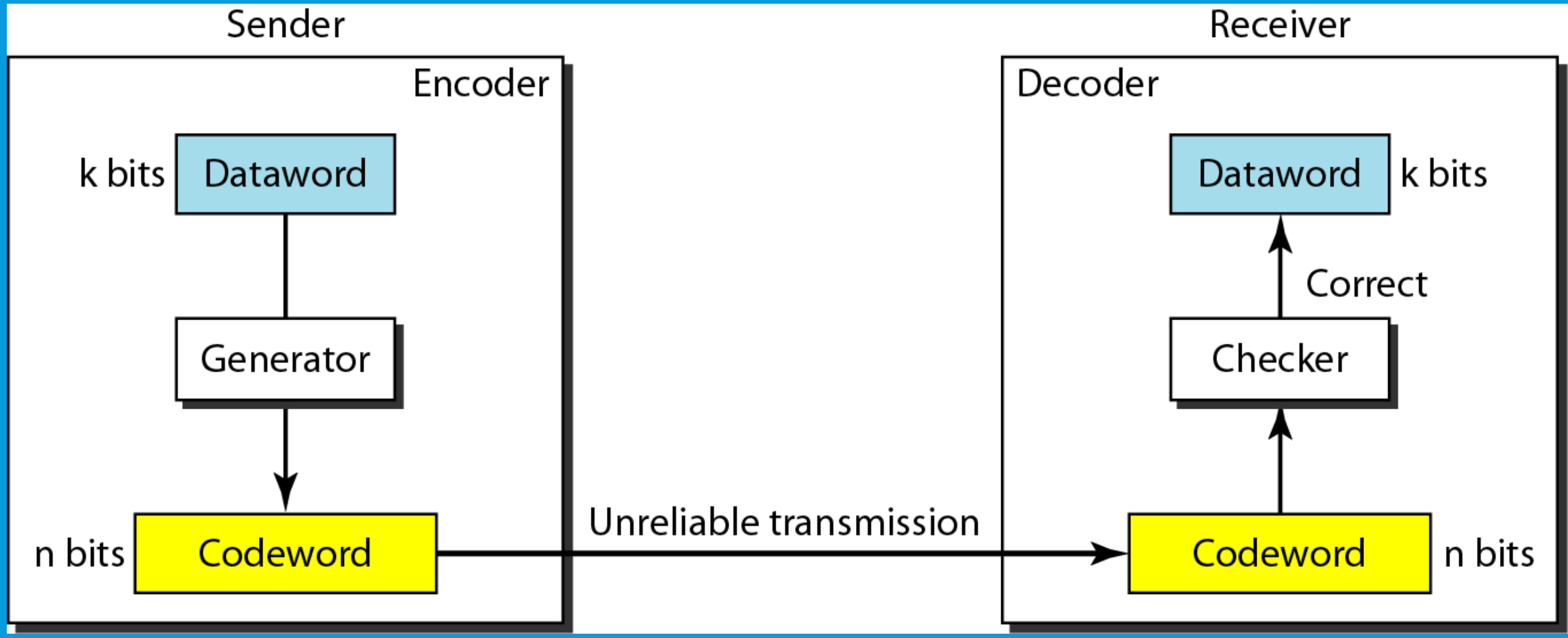
Datawords	*Codewords*
00	000
01	011
10	101
11	110

2. *The codeword is corrupted during transmission, and 111 is received. This is not a valid codeword and is discarded.*
3. *The codeword is corrupted during transmission, and 000 is received. This is a valid codeword. The receiver incorrectly extracts the dataword 00. Two corrupted bits have made the error undetectable.*

Datawords	*Codewords*
00	000
01	011
10	101
11	110

Note

An error-detecting code can detect only the types of errors for which it is designed; other types of errors may remain undetected.



Structure of encoder and decoder in error correction

Let us add more redundant bits to previous Example to see if the receiver can correct an error without knowing what was actually sent. We add 3 redundant bits to the 2-bit dataword to make 5-bit codewords. Table shows the datawords and codewords.

Dataword	*Codeword*
00	00000
01	01011
10	10101
11	11110

Assume the dataword is 01. The sender creates the codeword 01011. The codeword is corrupted during transmission, and 01001 is received. First, the receiver finds that the received codeword is not in the table. This means an error has occurred. The receiver, assuming that there is only 1 bit corrupted, uses the following strategy to guess the correct dataword.

Dataword	*Codeword*
00	00000
01	01011
10	10101
11	11110

1. *Comparing the received codeword with the first codeword in the table (01001 versus 00000), the receiver decides that the first codeword is not the one that was sent because there are two different bits.*
2. *By the same reasoning, the original codeword cannot be the third or fourth one in the table.*
3. *The original codeword must be the second one in the table because this is the only one that differs from the received codeword by 1 bit. The receiver replaces 01001 with 01011 and consults the table to find the dataword 01.*

Dataword	*Codeword*
00	00000
01	01011
10	10101
11	11110

Note

The Hamming distance between two words is the number of differences between corresponding bits.

Let us find the Hamming distance between two pairs of words.

1. The Hamming distance d(000, 011) is 2 because

$$000 \oplus 011 \text{ is } 011 \text{ (two 1s)}$$

2. The Hamming distance d(10101, 11110) is 3 because

$$10101 \oplus 11110 \text{ is } 01011 \text{ (three 1s)}$$

Note

The minimum Hamming distance is the smallest Hamming distance between all possible pairs in a set of words.

Find the minimum Hamming distance of the coding scheme in Table.

Solution

We first find all Hamming distances.

$d(000, 011) = 2$ $d(000, 101) = 2$ $d(000, 110) = 2$ $d(011, 101) = 2$
$d(011, 110) = 2$ $d(101, 110) = 2$

The d_{min} in this case is 2.

Datawords	*Codewords*
00	000
01	011
10	101
11	110

Find the minimum Hamming distance of the coding scheme in Table.

Solution

We first find all the Hamming distances.

$d(00000, 01011) = 3$	$d(00000, 10101) = 3$	$d(00000, 11110) = 4$
$d(01011, 10101) = 4$	$d(01011, 11110) = 3$	$d(10101, 11110) = 3$

The d_{min} in this case is 3.

Dataword	*Codeword*
00	00000
01	01011
10	10101
11	11110

Note

To guarantee the detection of up to s errors in all cases, the minimum Hamming distance in a block code must be $d_{min} = s + 1$.

The minimum Hamming distance for our first code scheme (Error Detection Table) is 2. This code guarantees detection of only a single error. For example, if the third codeword (101) is sent and one error occurs, the received codeword does not match any valid codeword. If two errors occur, however, the received codeword may match a valid codeword and the errors are not detected.

Datawords	*Codewords*
00	000
01	011
10	101
11	110

Our second block code scheme (Error Correction Table) has $d_{min} = 3$. This code can detect up to two errors. Again, we see that when any of the valid codewords is sent, two errors create a codeword which is not in the table of valid codewords. The receiver cannot be fooled.

However, some combinations of three errors change a valid codeword to another valid codeword. The receiver accepts the received codeword and the errors are undetected.

Dataword	*Codeword*
00	00000
01	01011
10	10101
11	11110

Note

To guarantee **correction** of up to *t* errors in all cases, the minimum Hamming distance in a block code must be $d_{min} = 2t + 1$.

A code scheme has a Hamming distance $d_{min} = 4$. What is the error detection and correction capability of this scheme?

Solution

This code guarantees the detection of up to three errors (s = 3), but it can correct up to one error. In other words, if this code is used for error correction, part of its capability is wasted. Error correction codes need to have an odd minimum distance (3, 5, 7, . . .).